

ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

A8ov5UqDdHWUAr8J81BCdUdoewNAPaGAtyH7TvQTYPoM

Date: 4/30/2025

Compounds: Amoxicillin, Caffeine, Aspirin, Dextromethorphan

RESEARCH PURPOSES ONLY

This analysis is generated for research and educational purposes. The predictions and insights provided should not be considered medical advice.

New Compound Analysis

Name: Amoxispiraflurin

Formula: C19H25N5O6

Weight: 415.43

Drug-likeness: 0.75

Activity:

Broad-spectrum antibacterial with added analgesic and mild central stimulant properties.

Target Analysis

Penicillin-binding protein

Binding Affinity: -8.5

Essential for bacterial cell wall synthesis; high affinity can lead to effective antibacterial actions.

Cyclooxygenase (COX)

Binding Affinity: -7.8

Inhibition reduces inflammation and pain, but can cause gastrointestinal side effects.

Adenosine receptor

Binding Affinity: -6.5

Activation can lead to increased alertness and decreased fatigue; may interact with cardiovascular functions.

NMDA receptor (via Dextromethorphan metabolite)

Binding Affinity: -6.7

Role in cough suppression; possible neuroprotective properties in low doses.

Toxicity Profile

Risk Level: Moderate

Affected Systems: Liver, Stomach, Central Nervous System

Mitigation Suggestions:

- Usage of gastroprotective agents
 - Regular liver function tests
 - Dose adjustments based on renal function
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Pharmacokinetics

Absorption:

Rapid oral absorption with moderate bioavailability.

Distribution:

Wide distribution due to lipophilicity; moderate plasma protein binding.

Metabolism:

Hepatic metabolism primarily through cytochrome P450 system.

Excretion:

Renal excretion with minor biliary contribution.

Half-life: 6-8 hours

Detailed Analysis

Amoxispiraflurin, which utilizes structural motifs from amoxicillin for antibacterial efficacy, aspirin's anti-inflammatory properties, caffeine's mild stimulant activity, and dextromethorphan's neuroactive effects, represents a novel compound for treating infections with concurrent systemic symptoms. This compound displays polypharmacology by interacting with multiple targets: bacterial PBPs to inhibit cell wall synthesis, COX for lowering inflammation, adenosine receptors for cognitive enhancement, and glutamatergic pathways via NMDA receptors. Despite its advantages, concerns remain about potential hepatotoxicity due to the metabolic demands, mandating cautious dosage and administrative strategies.

Combine base compounds, simulate new drug possibilities with AI, and earn rewards through Solana blockchain. Join our community of researchers and innovators in shaping the future of pharmaceutical discovery.

<https://ALCHEMUS.ai>